PENGZHEN LI

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EDUCATION

University of Tennessee, Knoxville expected 2027
Ph.D. in Economics
Committee: James Lake (Chair), Georg Schaur, Matthew Van Essen, Timothy Fitzgerald

Shandong University, Jinan 2018
M.A. in Finance — thesis published in *Technological Forecasting and Social Change* (2019); 1,000+ citations

Tongji University, Shanghai 2015
B.A. in Economics

RESEARCH INTERESTS

Environmental and Energy Economics, Agricultural and Resource Economics, Applied Econometrics, and International Trade.

PEER-REVIEWED PUBLICATIONS

Four SSCI/SCI-indexed publications; 1,132 total citations, h-index 4 (Google Scholar, August 2026).

"Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data." K. Du, **P. Li**, and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303. **1,000+ citations (Google Scholar).**

"Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability." **P. Li**, T. E. Yu, N. Labbé, N. Abdoulmoumine, M. Garcia-Perez, K. P. Hoyt, and B. C. English. *Biomass and Bioenergy*, 208 (2026), 108793.

"Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA." **P. Li**, T. E. Yu, C. Trejo-Pech, J. A. Larson, B. C. English, and D. N. Lanning. *Transportation Research Record*, 2678(9) (2024), 639–654.

"Beef price spread relationship with processing capacity utilization." C. Martinez, **P. Li**, C. N. Boyer, T. E. Yu, and J. G. Maples. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

WORKING PAPERS

Measured Quality Falls Within the Firm: Market Access and Chinese Exports [JMP]. [paper]
Abstract: This paper links the 2002 removal of tariff uncertainty on Chinese exports to the United States to a decline in measured quality within continuing export relationships. Comparing a firm's US and non-US shipments of the same product, measured quality falls by about 6 percent among cells that ship in both periods in the three years before an overlapping quota removal, and by about 11 percent in the full sample over the whole window, which contains that removal. Restricting the sample to cells that ship in both periods retains 83 percent of the full-window estimate and yields a decline twice as large as the full-sample estimate in the three clean years. The price and quantity

responses are not those of a markup cut on an unchanged good, and the decline is not the input-tariff upgrading of the same years, which varies only within firm and year and is absorbed by the design. We develop a model in which improved access can extend a seller's reach to buyers below a sorting boundary, who lower the return to appeal. Under stated conditions, appeal is capped and can fall. The continuing-cell price and quantity pattern is consistent with this mechanism but does not identify it against other sources of a demand-residual shift.

Trade Restrictiveness Indices with Discriminatory Tariffs with James Lake.
Abstract: A Trade Restrictiveness Index (TRI) is the uniform tariff that is welfare-equivalent to a country's actual tariff structure. Its standard implementation assigns a single tariff to each product, yet preferential trade agreements make tariffs on the same product differ across exporters, distorting where a country buys as well as what it buys. In an Armington model, we derive a TRI that separates these two distortions: for a small country, the squared index is, to second order, the square of the conventional index evaluated at mean product tariffs plus a term in the variance of tariffs across each product's exporters. We also derive the index for large importers, whose tariffs affect world prices. Across 123 importers over 2000–2020, the dispersion term accounts for 24 percent of the median importer's squared TRI in 2018, up from zero in the early 2000s. Preferences therefore reduce the TRI far less than they reduce average tariffs: in 2018 they cut the median importer's simple average tariff by 16 percent but its TRI by only 5 percent.

Price Risk, Irreversible Investment, and the Welfare Benchmark for the Section 45Z Clean Fuel Production Credit with T. Edward Yu. [paper] [slides]

How Should 45Z Pay for Risk? A Welfare-CVaR and Project-Finance Decomposition of the IRA Clean Fuel Production Credit with T. Edward Yu. [paper] [slides]

Estimating Pork Primal Price Relationships Using a Threshold Vector Autoregressive Analysis with Charles Martinez, Christopher N. Boyer, T. Edward Yu, and David Anderson. *SSRN working paper 4121953 (2022).* [paper]

Estimating Mean-Field Games with a Payoff Threshold: Candidates and Party Lists in Brazil with Matt Van Essen and Luiz Lima. *2024 J. Fred and Wilma Holly Award (best second-year paper).*
Abstract: Estimating a discrete-choice game of incomplete information means evaluating the joint action profile, which forecloses the settings such games describe: a Brazilian municipality fielding fifty-five council candidates across five lists has far too many action profiles to enumerate. We change the equilibrium concept rather than the algorithm. Anonymity makes the fixed point one dimension per list and none per candidate, and a discontinuous winner rule turns existence into a regime-consistency problem over proposed clearing sets, which we show can have no solution. Because a threshold pays several alternatives at once, two can be compared inside one market, and three log-odds contrasts identify the parameters with no probability of abstention anywhere in the estimator. At municipal scale the likelihood attenuates the coefficient on the anticipated chance of a seat, because a list's share is counted from a few dozen candidates, and a split-sample instrument on an unbiased reciprocal of that share recovers nine-tenths of it in simulation. On 926,900 Brazilian council candidacies the corrected coefficient is positive for non-incumbents and for all candidates, for every one of thirty-two demographic types, and with and without a control for local party strength; its magnitude under that control is not stable. Candidates with a university degree respond more to the chance of a seat than candidates without one, a difference found among non-incumbents and not among incumbents.

WORK IN PROGRESS

A Weight-Free Benchmark for Multilateral Tariff Bargaining with Georg Schaur. *Draft, August 2026.* [paper] [slides]

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Tennessee 2021 – 2023

USDA NIFA (two projects) and FAA Office of Environment and Energy (ASCENT) funded research on forest-residue biofuel and sustainable aviation fuel (SAF) supply chains in the Southeast U.S.

- Designed cost-minimizing and profit-maximizing supply-chain networks with mixed-integer programming (GAMS); results published in *Biomass and Bioenergy* and *Transportation Research Record*.
- Facility-siting and investment analysis combining econometric models with the IMPLAN regional impact platform; efficiency analysis of facilities and airports with Data Envelopment Analysis.
- Sensitivity and scenario analysis of feedstock quality and market conditions with extended mathematical programming (EMP), supervised machine learning, and @RISK.
- Managed high-resolution geospatial data in ArcGIS, R, and Python.

Other research experience

- *Panel and causal methods:* synthetic-control estimate of the long-run output effect of the 2008 Wenchuan earthquake; GMM estimates of MFP/CFAP payments on U.S. agricultural lending; Hansen panel-threshold models of green innovation and CO_2; mediation analysis of digital finance and emissions for Chinese listed firms.
- *Time series:* threshold VAR of regime switching in commodity-price linkages; VECM and multivariate GARCH of wholesale commodity-price volatility.
- *Machine learning:* drivers of CO_2 emissions from patent data; equity-premium prediction from *NYT* and *WSJ* text.

PRESENTATIONS

Western Regional Science Association, Annual Meeting, Hawaii 2023
Western Agricultural Economics Association, Annual Meeting, Santa Fe, NM 2022
Southern Agricultural Economics Association, Annual Meeting, New Orleans, LA 2022
Midwest Economic Theory and International Trade Meetings, Virginia Tech Spring 2025
University of Tennessee, Economics Brown Bag Seminar Series Spring 2024, Fall 2024
University of Tennessee, Applied Micro Group Spring 2024, Fall 2024

TEACHING EXPERIENCE

University of Tennessee

Instructor

ECON 351: Monetary Economics Spring 2025, Spring 2026
ECON 322: The Global Economy: Trade and Development Summer 2025
ECON 313: Intermediate Macroeconomics Summer 2026

Teaching Assistant

Ph.D. courses:
ECON 511: Microeconomic Theory Fall 2023, Fall 2024, Fall 2025
ECON 512: Advanced Microeconomics Spring 2024
ECON 581: Mathematical Methods in Economics Summer 2024
Master's courses:
AREC 525: Agribusiness Operations Research Methods Spring 2022, Spring 2023

HONORS AND AWARDS

Charles B. Garrison Award for Excellence in Teaching (best teaching), University of Tennessee	2026
Summer Graduate Assistantship, University of Tennessee	2025
J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee	2024
J. Fred and Wilma Holly Fellowship, University of Tennessee	2023 – present
Graduate Assistantship, University of Tennessee	2021 – present
Academic Scholarship for Master's Students, Shandong University	2017
Freshman Scholarship, Shandong University	2016
Outstanding Rainbow Volunteer, Tongji University (tutoring pupils in Shanghai)	2013 – 2015
National Endeavor Scholarship, Tongji University	2014
Tongji Scholarship of Excellence, Third Prize, Tongji University	2014

SKILLS

Programming Languages: Stata, R, Python, SQL, GAMS, MATLAB, LaTeX, ArcGIS, IMPLAN.
Methods: panel and causal econometrics (FE, IV, DiD, event study, SCM, GMM, panel threshold), time series (TVAR, VECM, mGARCH), mixed-integer and stochastic programming, DEA, cooperative and mean-field games, machine learning.
Credentials: CFA Level II.
Languages: Chinese (native), English (fluent).

REFERENCES

T. Edward Yu
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University of Tennessee
Email: tyu1@tennessee.edu

Georg Schaur
Leigh Burch III Economics Professor and Haslam Family Faculty Fellow
Department of Economics
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Associate Professor of Economics
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